A Growth-Based Method for Afforestation Carbon Dioxide Removal in PyPSA-Eur

Spatially and Temporally Resolved Sequestration Rates from Forest Yield Tables

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0.1 Afforestation as a Carbon Dioxide Removal Technology

Afforestation was first introduced in PyPSA-Eur by Fernandes et al. (2026) as a spatially resolved carbon dioxide removal (CDR) technology, deriving the land available for afforestation in each network node from the CORINE Land Cover (CLC) dataset. The present work expands that framework with a growth-based estimation of the carbon sequestration rate that is resolved both spatially (at NUTS-2 level) and temporally (monthly), and with a cost formulation that follows the local forest rotation age. The land-area input is taken unchanged from Fernandes et al. (2026); this document focuses on the sequestration rates and their temporal disaggregation.

The total sequestration potential of a network node is the product of two independent inputs:

$$\Pi_r = A_r \times \overline{\mathrm{MAI}}_{\mathrm{CO_2},r} \quad [\mathrm{tCO_2\,yr^{-1}}] \tag{1}$$

where A_r [ha] is the afforestation land area in node r and $\overline{\mathrm{MAI}}_{\mathrm{CO_2},r}$ [$\mathrm{tCO_2\,ha^{-1}\,yr^{-1}}$] is the node-level CO_2 sequestration rate derived below.

The land area A_r is derived, following Fernandes et al. (2026), from the CORINE Land Cover (CLC) dataset using only the *transitional woodland-shrub* class (CLC code 324) without converting cropland,pasture or existing mature forest. Figure 1 maps the resulting available area aggregated to the network nodes; it concentrates in Iberia, the Western Balkans and parts of Scandinavia, where shrubland and regenerating woodland are most extensive. The total land allocated to afforestation in the model amounts to approximately 15 Mha.

Key aspects. The method estimates, for a newly established forest in each region, the mean annual increment (MAI) of carbon over the optimal rotation, the optimal rotation age, and the carbon density reached at that age. The underlying assumption, consistent with EU legislation, is that beyond the optimal rotation age the forest carbon stock is maintained at least at that density level.

- **Potential.** The node potential (Eq. 1) uses the rotation-averaged MAI of a new forest.
- **EU compliance.** Beyond the optimal rotation age the stock is kept constant, leaving a margin for disturbances and minor harvest; the credited fraction of gross growth is set to 80 %.
- **Cost.** The per-hectare establishment cost from literature is annualised over the local optimal rotation age rather than a fixed lifetime, so that growth rate directly shapes the unit CDR cost.

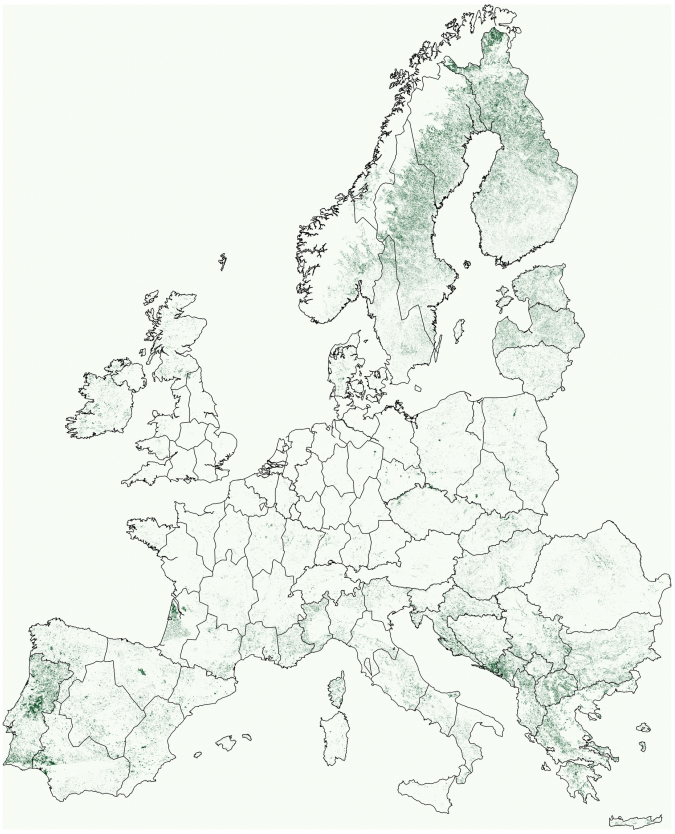


Figure 1: Land available for afforestation (green), from the CORINE Land Cover "transitional woodland-shrub" class (CLC code 324), aggregated to the 90-node network. Land-area derivation follows Fernandes et al. (2026).

0.1.1 Forest Yield Data: the Pilli et al. Library

Dataset. Sequestration rates are derived from the JRC forest growth library compiled by Pilli et al. (2024a) and distributed via Zenodo (Pilli et al., 2024b). It provides age-class-resolved growth curves for 222 forest types across 25 EU Member States (EU-27 excluding Cyprus and Malta), calibrated against national forest inventories (NFI) with reference years spanning 1992–2018, most in the 2005–2015 period. The library is consistent with EU Carbon Budget Model accounting (Grassi et al., 2018). Three datasets are used:

- **Standing stock** [$\mathrm{m^3\,ha^{-1}}$]: net merchantable volume under bark by 10-year age class for even-aged stands.
- **Net Annual Increment** (NAI, [$\mathrm{m^3\,ha^{-1}\,yr^{-1}}$]): merchantable volume growth rate by age class, harmonised across countries with species-specific corrections for bark, branches, top and stump.
- **Biomass Conversion and Expansion Factors** (BCEF, [$\mathrm{t_{DM}\,m^{-3}}$]): age-resolved factors converting merchantable volume to total aboveground dry biomass via species-specific allometric equations (Boudewyn et al., 2007).

Each curve is classified by country, administrative region (mostly NUTS-2), forest type (leading species), management type, and management strategy (even-aged or uneven-aged).

Management type and thinning. Only productive, even-aged high-forest stands are retained (management types "H", "HP", "HS", "MAN"); coppice, non-productive and protected forests are excluded, as they do not represent the productive new plantations targeted by afforestation. The distinction between standing stock and NAI is central for managed stands subject to repeated thinning. Thinning is the periodic removal of part of the growing trees, so the *standing stock* in the forest at any moment is therefore only part of all the wood the stand has produced over its life and systematically understates the productive capacity of the stand. The *total production* $\mathrm{TP}(T) = \int_0^T \mathrm{NAI}(t)\,dt$, which accumulates all increment including thinned material, recovers the full productivity and yields silviculturally meaningful rotation ages (Section 0.1.2). The thinned volume itself is not tabulated separately by Pilli; it is recovered implicitly as the gap between total production and standing stock, $\mathrm{Th}(T) = \mathrm{TP}(T) - V(T)$. This is consistent with the Pilli definition of NAI as the gross annual increment minus natural mortality only — not minus fellings (Pilli et al., 2024a) — so that cumulative NAI captures all the merchantable wood a stand grows, including the volume later removed as intermediate fellings. The standing stock is a median 54 % of total production evaluated at the rotation age across all productive forest types, i.e. fellings account for roughly 30–45 % of all the wood a stand produces over its rotation. Strictly, this gap is total fellings (intermediate thinnings

plus any salvage of disturbance-killed wood) rather than commercial thinning alone; a minority of forest types show inconsistencies between the independently harmonised stock and increment series, so a representative median is reported rather than per-type values. For example, Scots pine in northern Sweden shows a monotonically declining standing-stock MAI from age 25 yr, because growth after each thinning is too slow to lift the cumulative average, whereas the NAI-based total production peaks near 95 yr, consistent with observed Swedish harvesting practice (80–100 yr). Total production, with only a share θ of the thinned volume counted (Eq. 2), is therefore used to locate the rotation age, while the credited carbon density is read from the standing stock at that age, since CRCF credits count only carbon physically remaining in the forest, not thinned material transferred to wood products.

Coniferous and broadleaved species. Forest types are grouped into coniferous and broadleaved species. Conifers typically reach their increment peak earlier and at higher volumetric rates in boreal and montane settings, while broadleaves grow more slowly but carry higher wood basic density (higher BCEF). These groups therefore yield different volumetric MAI, different optimal rotation ages, and different carbon densities at rotation. To avoid biasing the result toward the locally dominant group and to respect biodiversity guidance, group means are computed separately and then blended with a 50/50 split (Section 0.1.2, Eqs. 6–7). The same conifer/broadleaf split is applied to: the rotation age T^* (cost annualisation), the carbon density ρ^* (cost denominator), and the monthly disaggregation (Section 0.1.4).

0.1.2 Rotation-Averaged Sequestration Rate

For new afforestation we compute a *rotation-averaged* MAI representing the long-term average CO_2 uptake until the growth-phase is completed. This avoids the overestimation that peak-growth increment would cause and is consistent with the EU LULUCF accounting framework (European Parliament and Council, 2018, 2023). The computation proceeds in four steps for each forest type f, region r, and management type m.

Step 1 — Optimal rotation age. The Pilli curves describe commercially managed stands, whose total production includes the full volume removed by thinning. Afforestation for carbon removal is managed less intensively, so only a share θ of the thinned volume $\mathrm{Th}(T) = \mathrm{TP}(T) - V(T)$ is counted when locating the rotation age:

$$\mathrm{TP}_\theta(T) = V(T) + \theta \left(\mathrm{TP}(T) - V(T)\right), \qquad \theta = 0.35 \tag{2}$$

$\theta = 1$ recovers full commercial thinning ($\mathrm{TP}_\theta = \mathrm{TP}$) and $\theta = 0$ locates the rotation on the standing stock alone ($\mathrm{TP}_\theta = V$). The value $\theta = 0.35$ represents the partial, restoration-

type thinning expected for carbon-removal plantings. The rotation age T^* is the age that maximises the MAI of TP_θ, bounded by $T_{\min} = 25\,\mathrm{yr}$ and $T_{\max} = 100\,\mathrm{yr}$. This is the classical silvicultural rotation, where the Current Annual Increment (CAI) crosses the MAI from above:

$$\frac{d}{dT}\left(\frac{\mathrm{TP}_\theta(T)}{T}\right) = \frac{\mathrm{CAI}_\theta(T) - \mathrm{MAI}_\theta(T)}{T} = 0 \;\Leftrightarrow\; T^* = \arg\max_{T_{\min}\le T\le T_{\max}} \frac{\mathrm{TP}_\theta(T)}{T} \tag{3}$$

The thinning share θ only moves T^*; the credited carbon is always the standing stock at T^* (Step 2). Counting less of the thinned volume moves the peak of TP_θ/T to younger ages, where the standing-stock MAI $V(T^*)/T^*$ is higher: the mean NUTS-2 rate rises from 4.3 ($\theta = 1$) to $5.3\,\mathrm{tCO_2\,ha^{-1}\,yr^{-1}}$ ($\theta = 0.35$). The upper cap of 100 yr matches the upper limit of European commercial rotations and prevents artefacts where total production is still rising at the end of the data range (e.g. Portuguese maritime pine, which grows at a nearly constant rate with no interior peak).

Step 2 — Volumetric MAI.

$$\mathrm{MAI}_{\mathrm{vol}} = \frac{V(T^*)}{T^*} \quad [\mathrm{m^3\,ha^{-1}\,yr^{-1}}] \tag{4}$$

Step 3 — Conversion to $\mathrm{CO_2}$.

$$\mathrm{MAI}_{\mathrm{CO_2}} = \mathrm{MAI}_{\mathrm{vol}} \times \mathrm{BCEF}(T^*) \times C_f \times \frac{M_{\mathrm{CO_2}}}{M_C} \times (1+R) \quad [\mathrm{tCO_2\,ha^{-1}\,yr^{-1}}] \tag{5}$$

where $\mathrm{BCEF}(T^*)$ converts merchantable volume to aboveground biomass, $C_f = 0.5$ is the carbon fraction of dry biomass (IPCC, 2006), $M_{\mathrm{CO_2}}/M_C = 44/12$, and $R = 0.25$ is the root-to-shoot ratio for belowground biomass (IPCC, 2006; Mokany et al., 2006).

Step 4 — Species blending and biodiversity. For each NUTS-2 region an equal-weight mean of the rotation-averaged MAI is computed within each species group,

$$\overline{\mathrm{MAI}}^{\,g}_{\mathrm{CO_2},r} = \frac{1}{|F^g_r|}\sum_{f\in F^g_r} \mathrm{MAI}_{\mathrm{CO_2},f,r}, \qquad g \in \{\mathrm{con},\,\mathrm{broad}\} \tag{6}$$

where F^g_r is the set of productive forest types of group g in region r, and the two group means are blended with equal weight:

$$\overline{\mathrm{MAI}}_{\mathrm{CO_2},r} = \tfrac{1}{2}\,\overline{\mathrm{MAI}}^{\mathrm{con}}_{\mathrm{CO_2},r} + \tfrac{1}{2}\,\overline{\mathrm{MAI}}^{\mathrm{broad}}_{\mathrm{CO_2},r} \tag{7}$$

If only one group has data, its mean is used directly. The 50/50 split satisfies the EU Biodiversity Strategy 2030 (European Commission, 2020) and most member-state minimum broadleaf mandates (20–30 %), and is consistent with IPCC Good Practice Guid-

ance (IPCC, 2006). Equal weighting within each group is adopted in the absence of NFI species-area data at NUTS-2 resolution.

0.1.3 Spatial Coverage: Mapping to All NUTS-2 Regions

The model requires a rate for every NUTS-2 region, but the library covers only 23 countries at a mix of NUTS-2, NUTS-1 and national resolution. Gaps arise both for countries outside the library (e.g. Norway, Switzerland, Turkey, the Western Balkans) and for individual regions below the NFI resolution (small islands, city-states). Full coverage is obtained through a fallback cascade applied in order:

1. **Direct NUTS-2 match.**
2. **NUTS-1 propagation:** a three-character region code is applied to all child NUTS-2 regions (e.g. `DEA`→`DEA1`–`DEA5`).
3. **Country-level propagation:** a national value is applied to all NUTS-2 regions in that country.
4. **Distance-based neighbour mean:** the mean of all NUTS-2 regions within 100 km (iterated until stable). The threshold captures meaningful sea crossings (Dover ≈34 km, Irish Sea ≈70 km) while excluding long-range links such as Scotland–Norway (≈300 km).

The source of each assigned value is recorded for reproducibility (Alamia, 2026). Figure 2 maps which step of the cascade supplied the rate for each NUTS-2 region: most of continental Europe is covered by direct Pilli entries, while peripheral and non-EU regions rely on propagation or neighbour means.

0.1.4 Temporal Weighting: Monthly Sequestration Profiles

The annual rate can be disaggregated in monthly rates for better temporal correlation with the other sectors in PyPSA-eur. The disaggregation is using Gross Primary Production (GPP) as a proxy for phenology: GPP measures photosynthetic carbon assimilation directly and is near zero in winter across temperate and boreal regions.

What GPP represents. Gross Primary Production is the gross rate at which vegetation fixes atmospheric CO_2 into organic carbon by photosynthesis, before any of it is returned to the atmosphere by plant respiration. It is a carbon-mass flux per unit ground area per unit time, reported here in $\mathrm{gC\,m^{-2}\,day^{-1}}$ — grams of carbon assimilated over one square metre of land in one day:

$$[\mathrm{GPP}] = \frac{\text{mass of carbon}}{\text{area} \times \text{time}} = \mathrm{gC\,m^{-2}\,day^{-1}}. \tag{8}$$

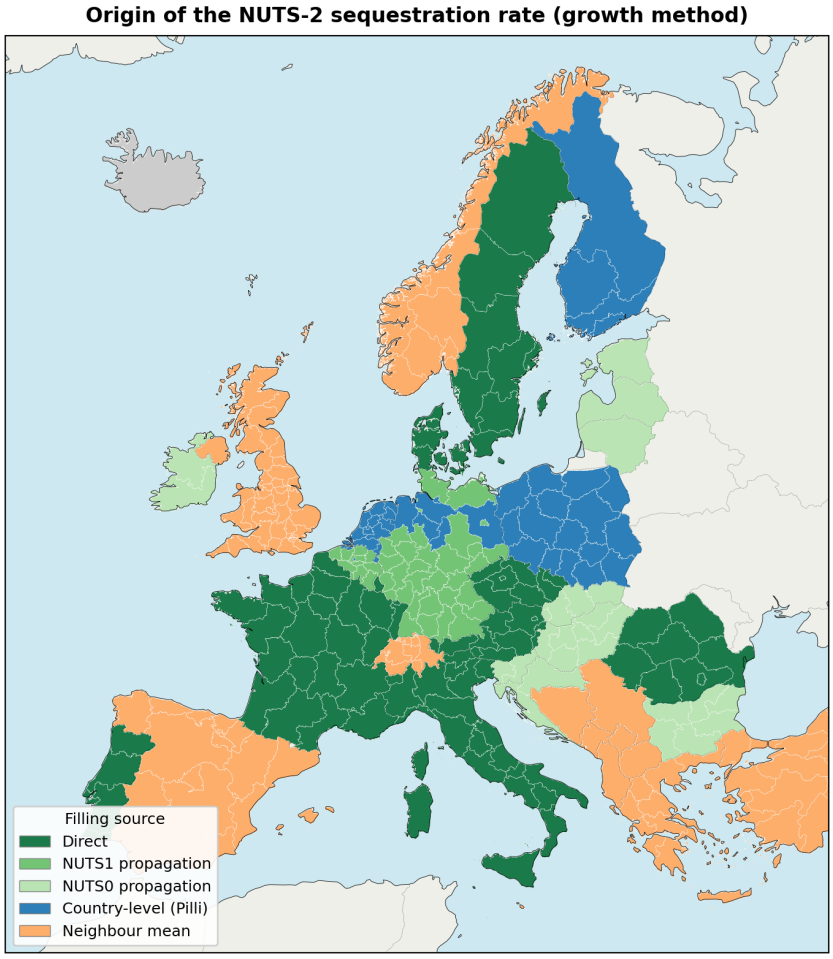


Figure 2: Origin of the NUTS-2 sequestration rate. Greens/blue are real Pilli data at decreasing resolution: direct NUTS-2 entry, NUTS-1/NUTS-0 propagation, and a country-level mean (for countries whose region codes do not resolve to NUTS-2, e.g. Finland and Poland). Orange marks regions without Pilli data, filled by distance-based neighbour mean (Section 0.1.3).

Because the monthly weights are formed as a ratio of two such fluxes (Eq. 9), the units and the absolute magnitude cancel: the weights retain only the *shape* of the seasonal cycle, which is all that is needed to redistribute the annual MAI across the year. GPP is used as a timing indicator only; its absolute level never enters the sequestration total.

Data and processing. Monthly weights are derived from the FluxCom RS+METEO ensemble (Jung et al., 2020), which provides daily global GPP at 0.5° resolution from a machine-learning upscaling of eddy-covariance towers, driven by ERA5 (for consistency with PyPSA-Eur's meteorology) and MODIS. Three years (2010–2012) are averaged into a climatological monthly field $G_m(\lambda, \phi)$. Each grid cell is assigned to a NUTS-2 region by point-in-polygon test on the NUTS-2013 boundaries (Eurostat, 2013), and the regional monthly mean $\bar{G}_{m,r}$ is taken over assigned cells with positive GPP (nearest-cell fallback for very small regions). The values are normalised to dimensionless weights summing to unity:

$$w_{m,r} = \frac{\bar{G}_{m,r}}{\sum_{m'=1}^{12} \bar{G}_{m',r}}, \qquad \sum_{m=1}^{12} w_{m,r} = 1 \tag{9}$$

Regions with no valid GPP fall back to the 100 km neighbour-mean profile, then the country-mean profile, then a uniform $w_{m,r} = 1/12$.

What the FluxCom RS+METEO product is. FluxCom does not measure GPP everywhere directly — it is a data-driven *upscaling*. Eddy-covariance towers measure the net CO_2 exchange between ecosystem and atmosphere at a few hundred sites worldwide, from which GPP is partitioned. A machine-learning model is trained to reproduce these tower fluxes from co-located predictors and then applied globally on a regular grid, yielding a gap-free field. The *RS+METEO* variant combines two predictor families: remote-sensing (RS) vegetation properties from MODIS satellite imagery (e.g. greenness and absorbed radiation) and gridded meteorological (METEO) drivers — here ERA5 reanalysis (temperature, radiation, etc.). The ERA5-driven variant is chosen so the seasonal climatology is consistent with the meteorological inputs used elsewhere in PyPSA-Eur.

Monthly rate.

$$\mathrm{Rate}_{m,r} = w_{m,r} \times \overline{\mathrm{MAI}}_{\mathrm{CO_2},r} \quad [\mathrm{tCO_2\,ha^{-1}\,month^{-1}}] \tag{10}$$

By construction $\sum_m \mathrm{Rate}_{m,r} = \overline{\mathrm{MAI}}_{\mathrm{CO_2},r}$, preserving the annual carbon budget. These weights are used in post-processing to remap the optimised annual CDR onto a physically meaningful seasonal profile (Section 0.1.6). Figure 3 shows the resulting monthly profiles by macro-region: northern and central Europe have a sharp summer peak and near-zero winter uptake, while southern Europe shows a broader, flatter season.

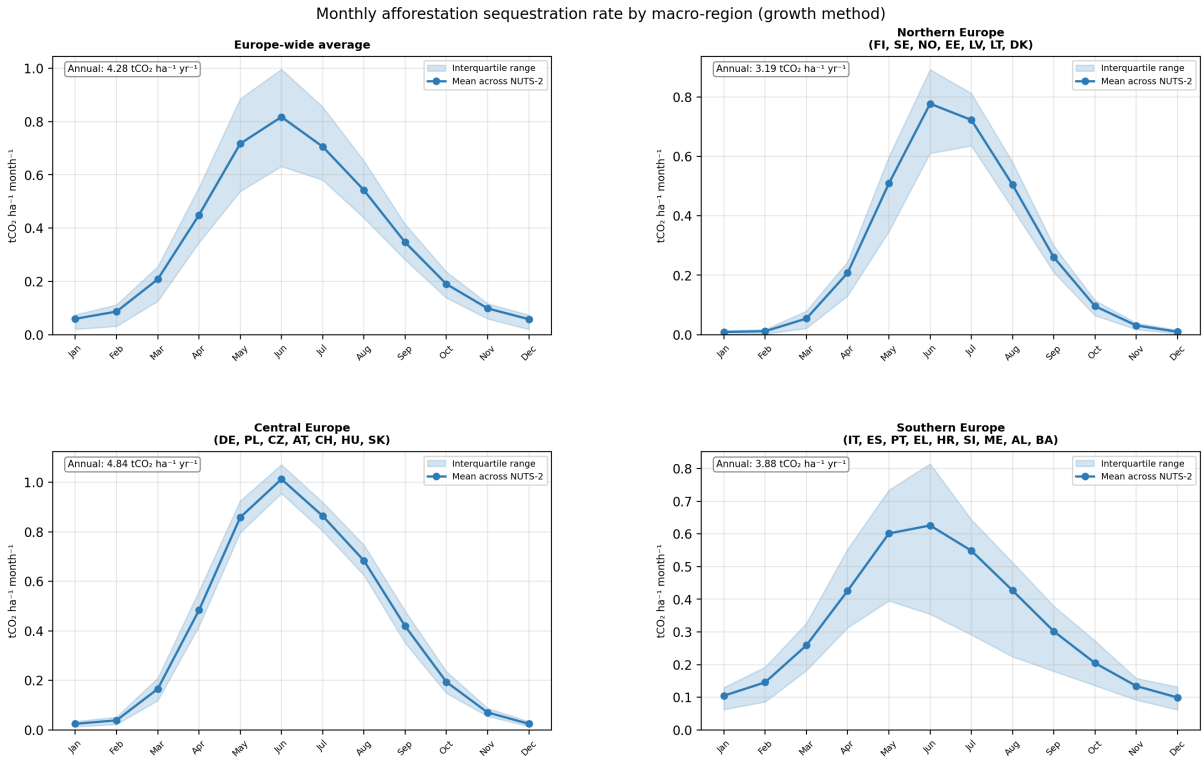


Figure 3: Monthly sequestration rate (mean ± interquartile range across NUTS-2 regions) by macro-region, from the GPP-weighted growth method.

0.1.5 Consistency with EU Policy and CRCF Crediting

The rotation-averaged MAI is consistent with the EU LULUCF Regulation (European Parliament and Council, 2018, 2023), which uses gross-net accounting of removals over a 20–30 year transition. Under the Carbon Removal Certification Framework (European Parliament and Council, 2024), the afforestation methodology (draft delegated act, 22 January 2026) defines a 30-year activity period followed by 40 years of monitoring — consistent with the PyPSA-Eur technology lifetime — and permits harvest provided trees are replanted with no net stock reduction, exactly the long-run average that the MAI represents.

Credits equal the net stock increase times $(1 - \mathrm{UNC})$, where UNC is a mandatory uncertainty deduction (min. 10 %), further reduced by a buffer-pool risk share (8–17 % for broadleaf/mixed forests). The net certified fraction is

$$\eta_{\mathrm{CRCF}} = (1 - \mathrm{UNC})(1 - r_{\mathrm{risk}}) \tag{11}$$

giving $\eta_{\mathrm{CRCF}} \approx 0.75$–$0.83$ for typical European conditions. The gross stock splits exactly into net certified stock, buffer reserve and a harvest/safety margin (the three sum to unity). In PyPSA-Eur η_{CRCF} is the `Link` efficiency, set via `afforestation[crcf_efficiency]` (default 0.80).

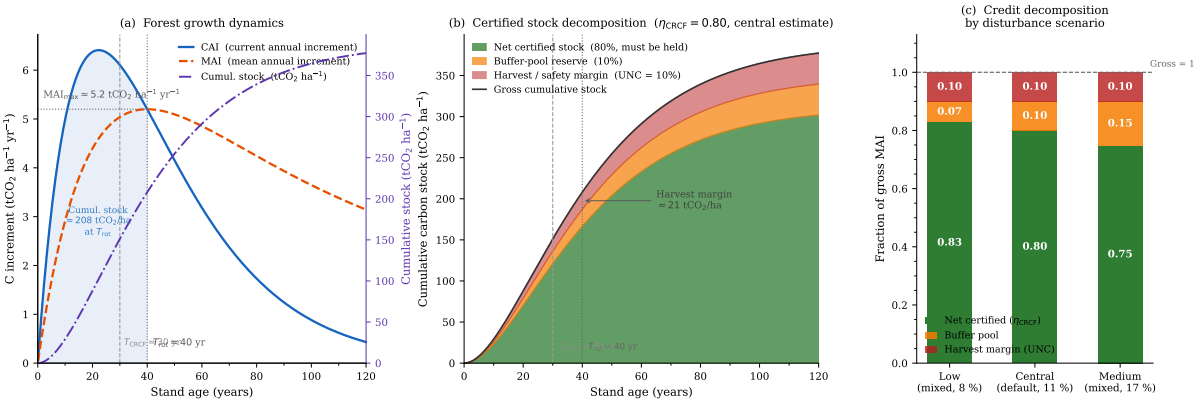


Figure 4: Forest growth dynamics and CRCF credit breakdown. *(a)* CAI and MAI for a representative European managed forest ($T_{\text{rot}} \approx 80\,\text{yr}$; EU median from Pilli et al. 2024b); dashed line: 30-yr CRCF activity period. *(b)* Gross cumulative stock split into net certified stock, buffer-pool reserve and harvest/safety margin ($\eta_{\text{CRCF}} = 0.80$, UNC$= 10\,\%$, $r = 11\,\%$). *(c)* The same fractions for three disturbance scenarios.

Summary of accounting and certification. Figure 5 ties the accounting together as three successive partitions, each normalised to its own base. First, the total wood produced over the rotation splits into the standing stock that remains in the forest (credited) and the thinnings removed along the way (counted elsewhere as harvested wood products). Second, the standing stem volume is expanded to total biomass by adding the non-merchantable aboveground components (branches, bark, top and stump, via the BCEF) and the belowground roots (via the root-to-shoot ratio). Third, the resulting gross CO_2 stock is partitioned by the CRCF into the certified credit, the buffer-pool reserve and the harvest/safety margin (Eq. 11). The three stages therefore operate on different quantities — volume, biomass mass and CO_2 — and should not be read as one split of a single "forest volume".

The same chain can be read as a carbon balance (Figure 6): starting from the merchantable stemwood a stand grows over its rotation, the thinnings are removed, the standing stem is expanded to total biomass by adding branches and roots, and the resulting gross stock is reduced by the CRCF buffer and safety margin to leave the certified credit.

0.1.6 PyPSA-Eur Implementation and Cost

Network representation. Each node r carries a `Store` on a dedicated `co2 afforestation` bus, supplied from `co2 atmosphere` through a one-way `Link` of efficiency η_{CRCF} (Eq. 11). The store `e_nom_max` $= \Pi_r$ (Eq. 1) is the annual CDR ceiling. The link `p_nom` is fixed to the seasonal peak rate at full deployment, with `p_min_pu` $= 0$ and `p_max_pu` $= 1$, so the optimiser allocates the annual removal $e_{\text{nom,opt},r} \leq \Pi_r$ freely across the year subject only to this instantaneous cap. Encoding the seasonal profile as time-varying `p_min_pu`(t)=`p_max_pu`(t) would prescribe the temporal shape exactly but densely pop-

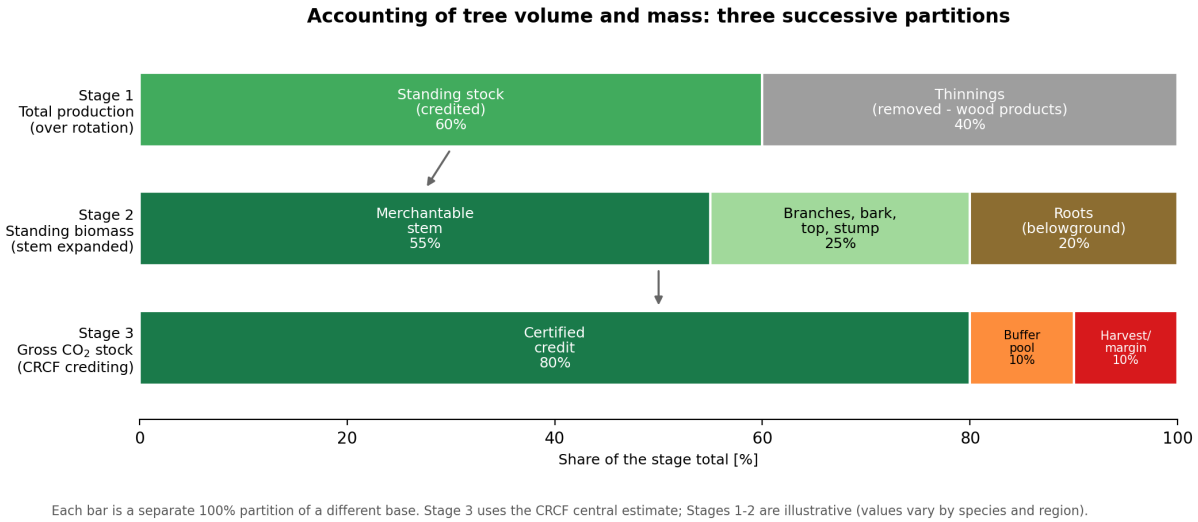


Figure 5: Tree volume and mass accounting as three successive partitions, each on its own 100 % base: total production (Stage 1), standing biomass (Stage 2) and gross CO_2 stock (Stage 3). Stage 3 uses the CRCF central estimate; Stages 1–2 are illustrative and vary by species and region.

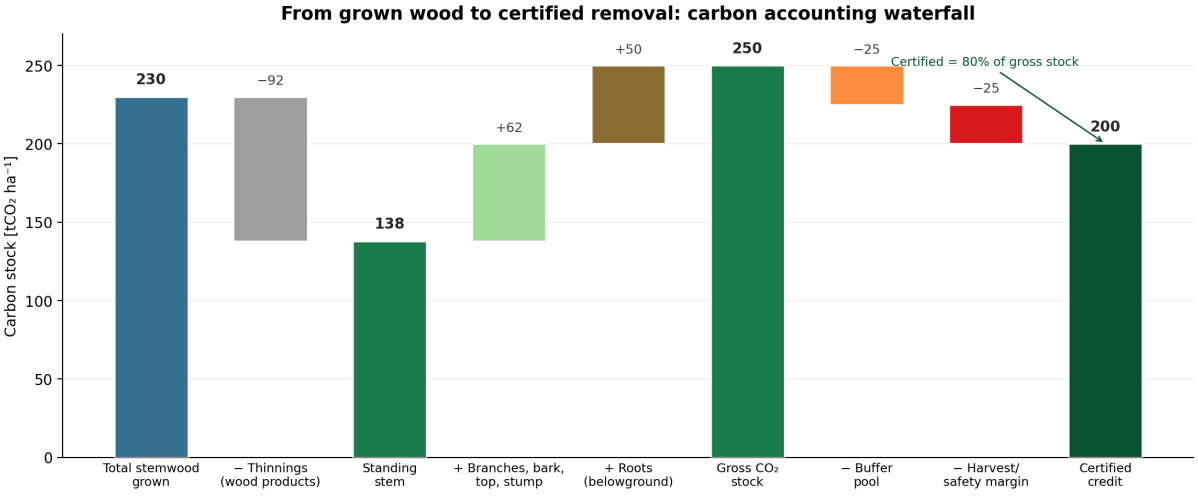


Figure 6: Carbon-balance waterfall from total merchantable stemwood grown to the certified removal credit (illustrative, $tCO_2\,ha^{-1}$ for a representative productive stand). Green/brown bars *add* mass (Stage 2 expansion), grey/orange/red bars *remove* it. *Assumptions:* the standing/thinnings split is taken from the Pilli data as the ratio of standing stock to cumulative NAI (total production) at T^* — here 60/40, within the observed range (median 54 % standing, interquartile range 43–76 % across 936 productive forest types); the standing stem is expanded to aboveground biomass via the BCEF and to total biomass via a root-to-shoot ratio $R = 0.25$, giving an illustrative gross-stock composition of 55/25/20 % stem/branches/roots; the gross CO_2 stock is split by the CRCF central estimate into 80 % certified, 10 % buffer pool and 10 % harvest/safety margin (Eq. 11). The absolute level scales with the regional carbon density ρ^* (Eq. 12).

ulates the LP ($\sim$8760 $\times$ N nonzeros vs. a single scalar) without adding detail to the results; capping `p_nom` at the seasonal peak avoids degenerate single-hour spikes while preserving the annual total. Before temporal plotting, `afforestation_postprocess.py` redistributes the link flows and state-of-charge per the GPP weights; all yearly totals are unchanged.

Node-specific cost. The carbon stock accumulated at the MAI optimum defines the carbon density at rotation,

$$\rho_r^* = \overline{\mathrm{MAI}}_{\mathrm{CO_2},\,r} \times T_r^* \quad [\mathrm{tCO_2\,ha^{-1}}], \tag{12}$$

directly computable from the Pilli data; both T_r^* and ρ_r^* are carried through the NUTS-2 to network-node overlay as area-weighted averages. The per-hectare establishment cost I is annualised over the local rotation age. A node-specific annuity factor $\mathrm{a}(r_d, T_r^*)$ is applied, with r_d the project discount rate:

$$\mathrm{capital_cost}_r = \frac{I \times \left[\mathrm{a}(r_d, T_r^*) + \mathrm{FOM}/100\right]}{\overline{\mathrm{MAI}}_{\mathrm{CO_2},\,r}} \quad [\mathrm{EUR\,tCO_2^{-1}}] \tag{13}$$

Longer-rotation nodes (e.g. boreal Scandinavia) thus face a higher annualised cost, reflecting the opportunity cost of capital tied up over a century-long rotation. Setting $r_d = 0$ reduces the annuity to $1/T_r^*$, i.e. spreading the overnight investment evenly across the rotation. The cost varies across nodes through both T_r^* and $\overline{\mathrm{MAI}}_{\mathrm{CO_2},\,r}$.

Figure 7 shows the spatial distribution of these two driving quantities. The pattern is ecologically consistent: the longest rotations (85–95 yr) occur in boreal Scandinavia and Finland — slow-growing Scots pine and Norway spruce — and in Portugal, where maritime pine binds the 95 yr cap; central European temperate forests show 60–80 yr and deciduous-dominated western Europe 30–45 yr. The highest carbon densities ($\rho^* > 400$ $\mathrm{tCO_2\,ha^{-1}}$) appear in productive central European forests where moderate-to-high growth combines with intermediate rotation.

0.1.7 Limitations

The method cannot introduce species or management types absent from a region's data, which may under-estimate the potential (excluding productive species not currently recorded) or over-estimate it through the NUTS-2 propagation cascade. Biodiversity enters only through the conservative 50/50 conifer/broadleaf blend. The two dominant sources of uncertainty are the lack of yield data in some European regions and the estimation of afforestable land from CLC.

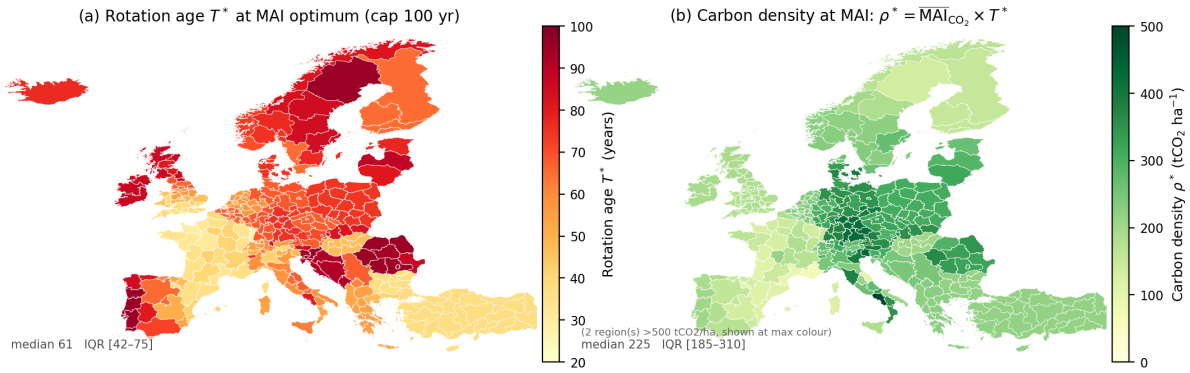


Figure 7: (a) Optimal rotation age T^* (capped at 100 yr) and (b) carbon density at rotation $\rho^* = \overline{\mathrm{MAI}}_{\mathrm{CO_2}} \times T^*$ across NUTS-2 regions, from the Pilli library (Pilli et al., 2024b) after conifer/broadleaf blending. Grey regions use fallback values.

0.1.8 Results

The rotation-averaged sequestration rates range from 1.9 to 10.8 $\mathrm{tCO_2\,ha^{-1}\,yr^{-1}}$ across the 122 regions of the Pilli library (NUTS-2, NUTS-1 or national), with a European median of 5.0 and mean of 5.3 $\mathrm{tCO_2\,ha^{-1}\,yr^{-1}}$ — within the 5–15 $\mathrm{tCO_2\,ha^{-1}\,yr^{-1}}$ range reported for temperate European afforestation (Avitabile et al., 2024; Nabuurs et al., 2013). As a validation case, the five Danish NUTS-2 regions yield 3.7–5.8 $\mathrm{tCO_2/ha/yr}$ (national average $\approx$5.0 $\mathrm{tCO_2/ha/yr}$), in reasonable agreement with literature estimates for a conifer/broadleaf mix; the equal-weight species averaging used here is slightly more conservative than an area-weighted approach favouring higher-productivity conifers. A selection of regional rates is given in Table 1.

The full spatial distribution of annual rates is shown in Figure 8: rates are highest across central Europe and lowest in boreal Scandinavia, where slow growth dominates despite long rotations.

Figure 9 shows the monthly sequestration rates aggregated to the PyPSA-Eur network. Rates are near zero in winter (December–February), rise sharply in spring, peak in June–July for temperate and boreal nodes, and decline in autumn. Southern nodes (Iberia, Italy, Greece) show a broader, flatter active season, reflecting year-round photosynthesis in Mediterranean climates; the highest monthly rates ($\approx$1.5 $\mathrm{tCO2ha^{-1}month^{-1}}$) occur in central European nodes at peak summer.

At the system level, a PyPSA-Eur run (90 nodes, 2050, 168-h resolution) gives a total afforestation potential of 75.0 $\mathrm{MtCO_2\,yr^{-1}}$, fully deployed by the optimiser at a system $\mathrm{CO_2}$ price of $\approx$476 EUR $\mathrm{tCO_2^{-1}}$ and an area-weighted unit cost of $\approx$205 EUR $\mathrm{tCO_2^{-1}}$. The node-level distribution of this potential (Figure 10) concentrates in Finland and Sweden, where large afforestable areas combine with substantial per-hectare rates. The largest national contributions (Figure 11) come from Finland, Sweden, Spain, Greece and Italy, reflecting the combination of available land and regional growth rates.

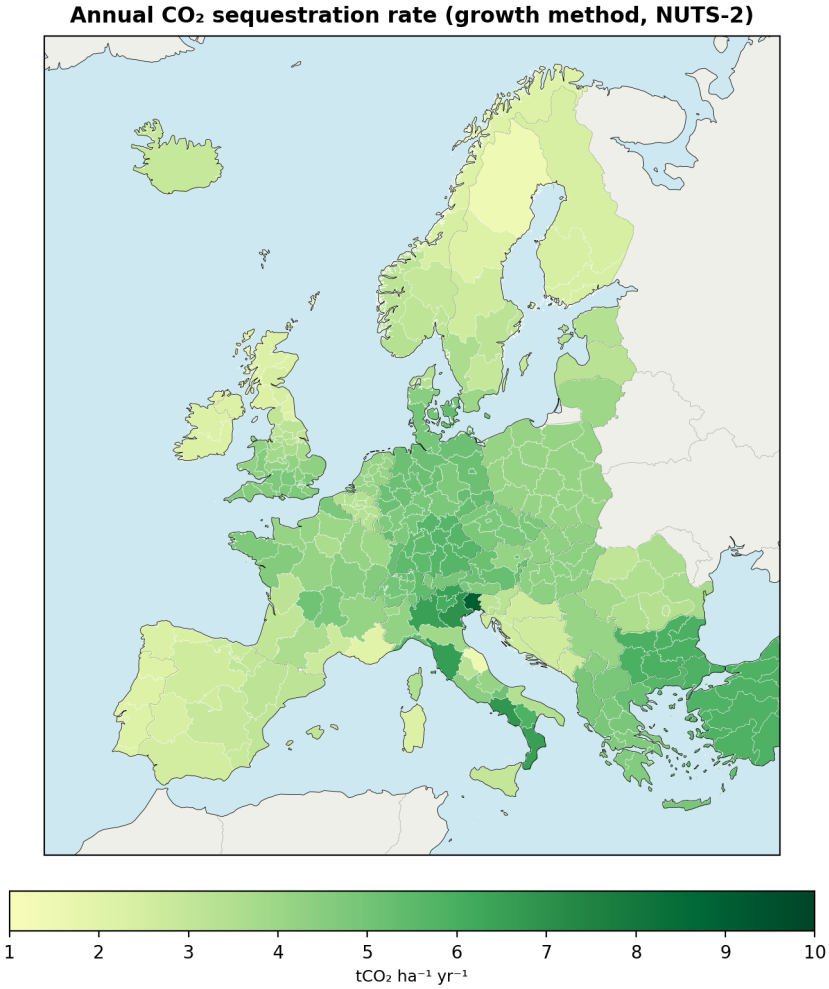


Figure 8: Annual rotation-averaged CO_2 sequestration rate [$tCO_2\,ha^{-1}\,yr^{-1}$] across NUTS-2 regions (growth method).

Table 1: Rotation-averaged CO_2 sequestration rates for selected NUTS-2 regions. $\bar{T}$: mean rotation age across productive forest types; n: number of productive types in the 50/50 con/broad blend. Dashes indicate regions filled via the spatial fallback cascade (Section 0.1.3).

Country	NUTS-2	Rate [$tCO_2\,ha^{-1}\,yr^{-1}$]	$\bar{T}$ [yr]	n
Austria	AT21	6.5	57	8
Belgium	BE35	6.1	—	—
Czechia	CZ01	4.9	62	4
Denmark	DK01	5.6	59	7
Denmark	DK05	3.7	64	7
Finland	FI1D	3.4	—	—
France	FR42	5.4	28	6
Germany	DE21	6.7	—	—
Ireland	IE01	7.0	—	—
Italy	ITH5	5.3	36	14
Poland	PL42	5.4	—	—
Romania	RO21	5.7	43	9
Sweden	SE11	4.0	62	4
Sweden	SE33	1.9	58	4

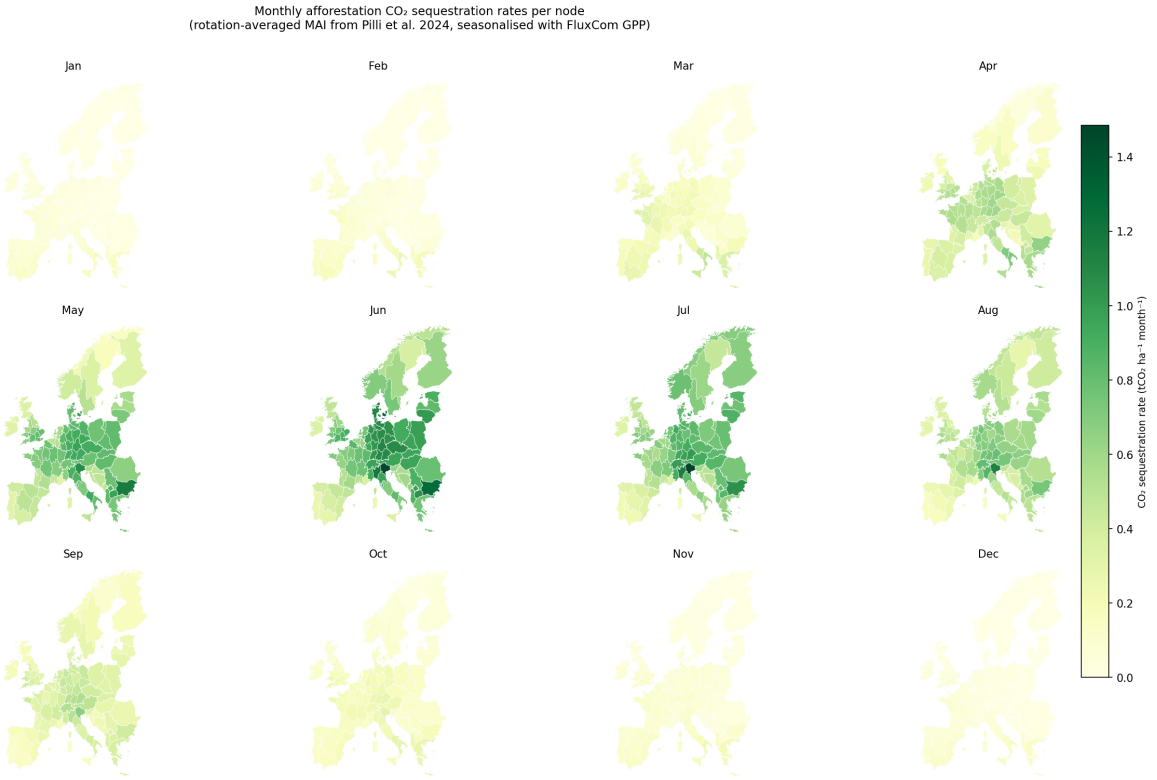


Figure 9: Monthly afforestation CO_2 sequestration rates per node [$tCO2ha^{-1}month^{-1}$] for the 90-node network, from the product of the rotation-averaged MAI and the GPP-derived monthly weight $w_{m,r}$ (Eq. 10).

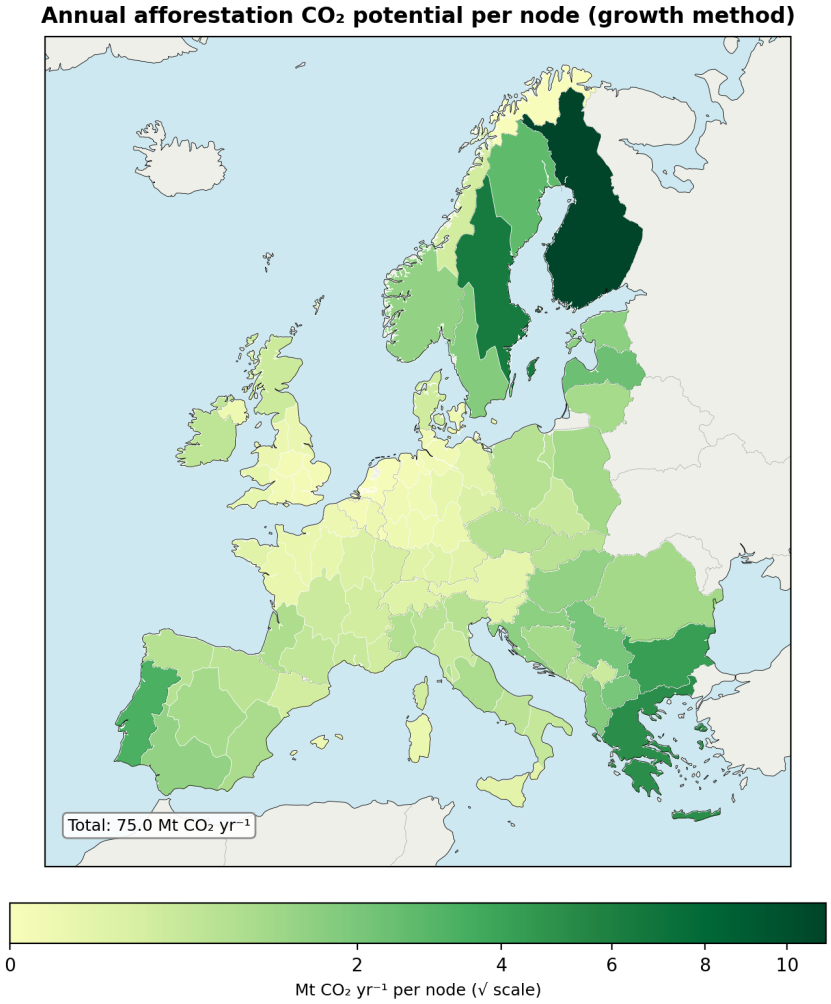


Figure 10: Annual afforestation CO_2 potential per network node [$MtCO_2\,yr^{-1}$], growth method (90-node, 2050). Square-root colour scale; total 75.0 $MtCO_2\,yr^{-1}$.

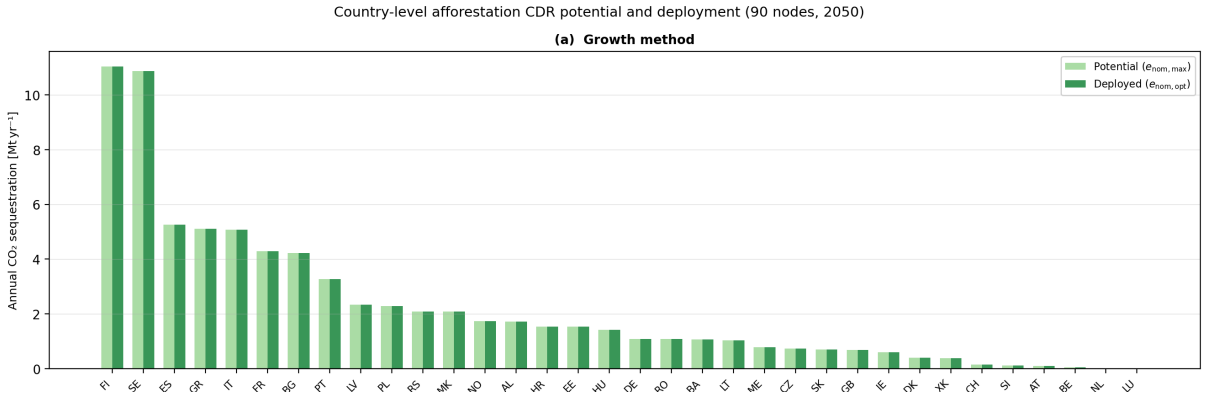


Figure 11: Country-level afforestation potential ($e_{\mathrm{nom,max}}$) and deployed CDR ($e_{\mathrm{nom,opt}}$) [$MtCO_2\,yr^{-1}$], 90-node 2050 run. Potential is fully deployed in all countries.

Over the annual horizon, the `co2 afforestation` component follows the expected seasonal pattern (Figure 12), peaking in spring and summer and falling in winter, in line with the GPP-derived weights recovered in post-processing.

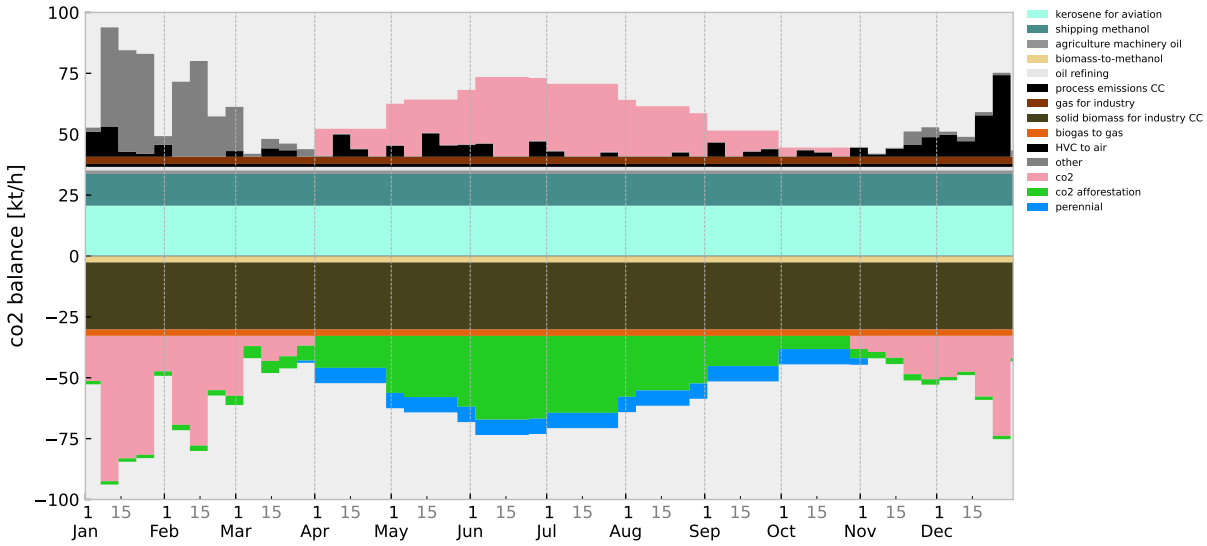


Figure 12: Annual CO_2 bus balance (90-node, 2050, 168-h). The `co2 afforestation` component (green) shows the seasonal signal recovered by `afforestation_postprocess.py`.

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